attempts by physicists such as Hawking and James Hartle of the University of California at Santa Barbara have tried to bring the question of cosmological initial conditions within the umbrella of physical theory, but all such attempts remain inconclusive. In the context of string/M-theory, our cosmological understanding is, at present, just too primitive to determine whether our candidate “theory of everything” truly lives up to its name and determines its own cosmological initial conditions, thereby elevating them to the status of physical law. This is a prime question for future research.

But even beyond the issue of initial conditions and their impact on the ensuing historical twists and turns of cosmic evolution, some recent and highly speculative proposals have argued for yet other potential limits on the explanatory power of any final theory. No one knows if these ideas are right or wrong, and certainly they currently lie on the outskirts of mainstream science. But they do highlight—albeit in a rather provocative and speculative manner—an obstacle that any proposed final theory may encounter.

The basic idea rests upon the following possibility. Imagine that what we call the universe is actually only one tiny part of a vastly larger cosmological expanse, one of an enormous number of island universes scattered across a grand cosmological archipelago. Although this might sound rather far-fetched—and in the end it may well be—André Linde has suggested a concrete mechanism that might lead to such a gargantuan universe. Linde has found that the brief but crucial burst of inflationary expansion discussed earlier may not have been a unique, one-time event. Instead, he argues, the conditions for inflationary expansion may happen repeatedly in isolated regions peppered throughout the cosmos, which then undergo their own inflationary ballooning in size, evolving into new, separate universes. And in each of these universes, the process continues, with new universes sprouting from far-flung regions in the old, generating a never ending web of ballooning cosmic expanses. The terminology gets a little cumbersome, but let’s follow fashion and call this greatly expanded notion of the universe the multiverse, with each of the constituent parts being called a universe.

The central observation is that whereas in Chapter 7 we noted that everything we know points toward a consistent and uniform physics